SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I N. Chawran Aryan (192365053) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

N . Chawran Aryan
Signature of the student:

Annexure A

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Reactive Agent-Based Traffic Control System

Introduction

A **Reactive Agent** makes decisions based on the current environment. In a smart traffic control system, the agent continuously receives real-time traffic information and immediately changes traffic signals according to road conditions.

System Architecture

```
Traffic Cameras     Road Sensors      GPS / Vehicle Data
       |                |                 |
       +----------------+-----------------+
                        |
                        v
               Streaming Data Input
                        |
                        v
                 Reactive Agent
                        |
             +----------+----------+
             |                     |
             v                     v
       Traffic Analysis      Congestion Check
             |                     |
             +----------+----------+
                        |
                        v
                 Signal Controller
                        |
                        v
              Traffic Light Changes
```

Working

The system collects:

- Number of vehicles
- Traffic density
- Waiting time
- Emergency vehicle information
- Road congestion

The reactive agent follows condition-action rules:

```
IF traffic_density > threshold
THEN increase green signal time

IF emergency_vehicle_detected
THEN provide priority signal

IF road_is_empty
THEN reduce green signal duration
```

Streaming Data Processing

Real-time traffic data can be processed continuously using a distributed data processing system. The agent analyzes the latest traffic condition and immediately reacts without requiring a complex future plan.

Advantages

- Reduces traffic congestion
- Provides faster response
- Gives priority to emergency vehicles
- Automatically adapts signal timing
- Supports real-time decision-making

Conclusion

A Reactive Agent-based traffic control system uses current traffic information to make immediate decisions. Continuous data processing and automatic signal adjustment help minimize congestion.

2. Goal-Based Agent for Healthcare Diagnosis

Introduction

A **Goal-Based Agent** selects actions based on a specific objective. In a healthcare diagnosis system, the main goal is to identify the patient's disease accurately and recommend an appropriate treatment.

System Architecture

```
Patient Data
   |
   v
Data Collection
   |
   v
Data Preprocessing
   |
   v
Goal-Based Agent
   |
   +----------------------+
   |                 |
   v                  v
Diagnosis Goal      Treatment Goal
   |                 |
   +----------+----------+
            |
            v
     Decision-Making Engine
            |
            v
    Optimal Treatment Path
```

Agent Working

The agent receives:

- Symptoms
- Medical history
- Laboratory reports
- Vital signs
- Previous diagnoses

The agent sets a goal:
Goal → Identify the correct disease
↓
Analyze patient data
↓
Compare possible diseases
↓
Evaluate possible treatments
↓
Select the best treatment pathway

Decision-Making Logic

The agent evaluates multiple possible states and selects actions that move closer to the goal.
Example:
Symptoms: Fever + Cough + Breathing Difficulty

Possible Conditions:
1. Flu
2. Pneumonia
3. Other respiratory infection

Agent analyzes:
Symptoms + Test Results + Medical History

↓

Select most suitable diagnosis
↓
Recommend appropriate treatment pathway

Data Pipeline

Patient Data
↓
Collection
↓
Cleaning and Preprocessing
↓
Feature Extraction
↓
AI Diagnosis Model
↓
Goal-Based Agent
↓
Treatment Recommendation

Conclusion

The Goal-Based Agent evaluates patient information, compares different possibilities, and selects actions that help achieve the goal of accurate diagnosis and optimal treatment.

3. Utility-Based Agent for E-Commerce Recommendations

Introduction

A **Utility-Based Agent** selects the action that provides the highest utility or satisfaction. In an e-commerce recommendation system, the agent evaluates multiple products and recommends the products with the highest expected user satisfaction.

System Architecture

```
User Data + Product Data
        |
        v
   Large-Scale Data
     Processing
        |
        v
    Feature Analysis
        |
        v
     Utility Agent
        |
  +------+------+------+
  |          |     |
  v          v     v
Product A    Product B  Product C
  |          |     |
  +------+------+------+
        |
        v
  Calculate Utility Score
        |
        v
 Recommend Highest Score
```

Utility Calculation

The utility value may depend on:

- User interests
- Product rating
- Product price
- Previous purchases
- Product relevance
- Availability
- Customer feedback

Example:

Utility Score =

(Relevance × 0.4)
+ (Rating × 0.3)
+ (Price Preference × 0.2)
+ (Availability × 0.1)
The product with the highest utility score is recommended.

Large-Scale Data Processing

PySpark can process large volumes of:

- Customer interactions
- Product information
- Search history
- Purchase history
- Product ratings

RDD operations such as map(), filter(), reduceByKey(), and join() can be used to process and combine this information.

Agent Workflow

User Activity
↓
Analyze Preferences
↓
Generate Product Options
↓
Calculate Utility Values
↓
Compare Options
↓
Select Highest Utility
↓
Personalized Recommendation

Conclusion

The Utility-Based Agent evaluates several possible recommendations and selects the one with the highest utility value, thereby improving user satisfaction and personalization.

4. Learning Agent for Personalized Education System

Introduction

A **Learning Agent** improves its performance over time by learning from previous actions and feedback. In a personalized education system, the agent analyzes student performance and adapts learning content accordingly.

System Architecture

Student Activity
|
v
Learning Data Collection
|
v
PySpark Distributed Processing
|

```
     v
Learning Agent
     |
+-----+------+------+------+------+
|         |          |
v         v          v
Performance  Feedback   Content Analysis
|         |          |
+------------+-------------+
          |
          v
     Adaptive Learning
          |
          v
   Personalized Content
```

Components of Learning Agent

1. Performance Element

Selects learning content for the student.

2. Learning Element

Improves the system using previous performance and feedback.

3. Critic

Evaluates whether the recommended content improved student performance.

4. Problem Generator

Suggests new learning strategies or activities.

Adaptive Learning Process

Student completes lesson
↓
Performance is analyzed
↓
Feedback is collected
↓
Learning Agent updates knowledge
↓
Difficulty level is adjusted
↓
New personalized content is recommended

Example:

Low Score
↓
Provide Basic Learning Material

Medium Score
↓
Provide Intermediate Exercises

High Score
↓

Provide Advanced Content

PySpark Integration

For thousands of learners, student data can be distributed across multiple nodes.

RDD operations:

- map() – extract student performance data.
- filter() – identify weak or high-performing students.
- reduceByKey() – calculate scores.
- groupByKey() – group learning activities.
- cache() – improve repeated analysis.

Conclusion

The Learning Agent continuously improves using student feedback and performance data. PySpark provides scalable distributed processing for large numbers of students

5. Multi-Agent System for Disaster Management

Introduction

A **Multi-Agent System** consists of multiple intelligent agents that communicate and cooperate to achieve a common objective. In disaster management, different agents work together to detect disasters, analyze risks, send alerts, and coordinate emergency responses.

System Architecture

```
   Sensors / Cameras / Social Media
              |
              v
         Sensor Agents
              |
              v
         Shared Data Layer
              |
    +-----------+-----------+
    |                   |
    v                   v
 Decision Agents     Analysis Agents
    |                   |
    +-----------+-----------+
              |
              v
       Communication Agents
              |
              v
        Alert / Warning System
              |
              v
       Emergency Response Teams
```

Types of Agents

1. Sensor Agent

Collects real-time information from:

- Flood sensors
- Earthquake sensors
- Weather stations

- Cameras
- Satellite data

2. Analysis Agent
Processes collected information and identifies possible disaster patterns.

3. Decision Agent
Evaluates:
- Disaster severity
- Affected locations
- Population risk
- Required emergency resources

4. Communication Agent
Sends warnings to:
- Citizens
- Government authorities
- Hospitals
- Rescue teams

5. Response Agent
Coordinates emergency actions such as:
- Evacuation
- Rescue operations
- Resource allocation
- Route planning

Agent Coordination Strategy

Sensor Agent detects abnormal activity
↓
Data shared with Analysis Agent
↓
Analysis Agent verifies disaster
↓
Decision Agent determines severity
↓
Communication Agent sends warning
↓
Response Agent coordinates rescue

Data Sharing and Real-Time Decision Making
Agents communicate through a shared distributed data system. Real-time information is continuously updated, allowing agents to react quickly to changing disaster conditions.

Advantages
- Faster disaster detection
- Real-time communication
- Distributed decision-making
- Improved coordination
- Scalable architecture
- Better emergency response

Conclusion
A Multi-Agent Disaster Management System combines sensor, analysis, decision, communication, and response agents. By sharing information and making coordinated decisions, the system can provide faster

disaster detection and effective emergency response.